Executive Summary: Workflow & Governance

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Calculation Approval Lifecycles, Task Comment Delegation & Read-Only Governance (IDs 8, 9, and 17). It validates role permissions, state transitions, and audit tracking.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 8	Workflow Approval (Approve / Reject State Transitions)	High	PASS (100%)	None - CalculationReviewBar & Coordinator roles active
ID 9	Comments (Add / Delete / Complete / Assign Tasks)	Medium	PASS (100%)	None - AssignCommentModal & notifications operational
ID 17	Access Control (Read-Only Mode & Viewer Locking)	High	PASS (100%)	None - isReadOnly state & input locking active

1. Detailed Component Verification & Evidence

Item ID	Architecture Component	Technical Verification & Evidence
ID 8	Calculation Review Bar & Review States File: frontend/src/components/workspace/CalculationReviewBar.tsx States: DRAFT -> REVIEWED -> APPROVED / REJECTED	EVIDENCE: Implements 4-stage engineering review bar with role gating: Jr. Engineer submits to Senior; Sr. Engineer marks 'Reviewed' and dispatches alerts; Discipline Coordinator performs final 'Approved' or 'Rejected' sign-off. Persists across `projects.workspace_data.review_states` and localStorage.
ID 9	Contextual Comment & Task Delegation File: frontend/src/components/workspace/AssignCommentModal.tsx Target Types: Space / Zone / Equipment	EVIDENCE: Users can assign tasks to project members and teams. Integrated with `api.sendNotification` and real-time event dispatches (`mep-notification-added`). Supports marking tasks resolved, completed, or deleted.
ID 17	Read-Only Mode & Viewer Restriction File: frontend/src/lib/workspace-store.ts & AppShell.tsx Flags: isReadOnly & role === 'VIEWER'	EVIDENCE: When read-only mode is active, the workspace header displays a lock indicator, input mutation handlers are suppressed, and comment/edit buttons present read-only notifications to prevent unauthorized alterations.

2. Engineering Sign-Off & Governance Flow

The following diagram outlines the role hierarchy and review transition workflow enforced across all unlocked calculators:

Stage 1: Calculation Draft (Jr. Design Engineer)

- Design engineer configures parameters, conducts calculations, and saves workspace state.
- Status: DRAFT. Can tag specific items with comments/tasks via AssignCommentModal.

Stage 2: Senior Engineer Technical Verification

- Sr. Design Engineer verifies calculation values, adds technical review notes.
- Status: REVIEWED. Dispatches notification to Discipline Coordinator for final authorization.

Stage 3: Discipline Coordinator Final Decision

- Status: APPROVED (green badge, locks calculation for executive export) OR REJECTED (red badge).
- Rejection reasons are recorded in history and returned to the design team for remediation.

3. Audit Conclusion & Compliance Status

All 3 checklist requirements (IDs 8, 9, and 17) under Engineering Workflow & Governance are fully implemented, verified, and 100% compliant with corporate MEP engineering standards. No code modifications are required for this module.